Configuring and enabling policy-driven data protection

SnapManager Oracle

NetApp
August 06, 2021

This PDF was generated from https://docs.netapp.com/us-en/snapmanager-oracle/unix-administration/task_configuring_snapdrive_when_rbac_is_enabled.html on August 06, 2021. Always check docs.netapp.com for the latest.

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Configuring and enabling policy-driven data protection

You must configure SnapDrive and the DataFabric Manager server to enable data protection on the profile to protect backups on the secondary storage systems. You can select the protection policies in the Protection Manager's console to specify how database backups will be protected.

> You must ensure that OnCommand Unified Manager is installed on a separate server to enable data protection.

Configuring DataFabric Manager server and SnapDrive when RBAC is enabled

When role-based access control (RBAC) is enabled, you must configure the DataFabric Manager server to include the RBAC capabilities. You must also register the SnapDrive user created in the DataFabric Manager server and root user of the storage system in SnapDrive.

1. Configure the DataFabric Manager server.
   a. To refresh the DataFabric Manager server to update the changes made directly on the storage system by the target database, enter the following command:dfm host discover storage_system
   b. Create a new user in the DataFabric Manager server and set the password.
   c. To add the operating system user to the DataFabric Manager server administration list, enter the following command:dfm user add sd-admin
   d. To create a new role in the DataFabric Manager server, enter the following command:dfm role create sd-admin-role
   e. To add the DFM.Core.AccessCheck Global capability to the role, enter the following command:dfm role add sd-admin-role DFM.Core.AccessCheck Global
   f. To add sd-admin-role to the operating system user, enter the following command:dfm user role set sd-adminsd-admin-role
   g. To create another role in the DataFabric Manager server for the SnapDrive root user, enter the following command:dfm role create sd-protect
   h. To add RBAC capabilities to the role created for the SnapDrive root user or the administrator, enter the following commands:dfm role add sd-protect SD.Config.Read Globaldfm role add sd-protect SD.Config.Write Globaldfm role add sd-protect SD.Config.Delete Globaldfm role add sd-protect SD.Storage.Read Globaldfm role add sd-protect DFM.Database.Write Globaldfm role add sd-protect GlobalDataProtection
   i. To add the target database oracle user to the list of administrators in the DataFabric Manager server and assign the sd-protect role, enter the following command:dfm user add -r sd-protecttardb_host1\oracle
   j. To add the storage system used by the target database in the DataFabric Manager server, enter the following command:dfm host set storage_system hostLogin=oracle hostPassword=password
   k. To create a new role in the storage system used by the target database in the DataFabric Manager

server, enter the following command:dfm host role create -h storage_system-c "api-**,login-**" storage-rbac-role

l. To create a new group in the storage system and assign the new role created in the DataFabric Manager server, enter the following command:dfm host usergroup create -h storage_system-r storage-rbac-rolestorage-rbac-group

m. To create a new user in the storage system and assign the new role and the group created in the DataFabric Manager server, enter the following command:dfm host user create -h storage_system-r storage-rbac-role -p password -g storage-rbac-grouptardb_host1

2. Configure SnapDrive.

   a. To register the credentials of the sd-admin user with SnapDrive, enter the following command:snapdrive config set -dfm sd-admindfm_host

   b. To register the root user or the administrator of the storage system with SnapDrive, enter the following command:snapdrive config set tardb_host1storage_system

Configuring SnapDrive when RBAC is not enabled

You must register the root user or the administrator of the DataFabric Manager server and root user of the storage system with SnapDrive to enable data protection.

1. To refresh the DataFabric Manager server to update the changes made directly on the storage system by the target database, enter the following command:

   dfm host discover storage_system

2. To register the root user or the administrator of the DataFabric Manager server with SnapDrive, enter the following command:

   snapdrive config set -dfm Administratordfm_host

3. To Register the root user or the administrator of the storage system with SnapDrive, enter the following command:

   snapdrive config set root storage_system

Understanding enabling or disabling of data protection in profile

You can enable or disable data protection while creating or updating a database profile.

To create a protected backup of a database on the secondary storage resources, database administrators and storage administrators perform the following actions.

If you want to…	Then…
Create or edit a profile	To create or edit a profile, perform the following: • Enable backup protection to the secondary storage. • If you are using Data ONTAP operating in 7-Mode and have installed Protection Manager, you can select the policies created by the storage or backup administrator in Protection Manager. If you are using Data ONTAP operating in 7-Mode and protection is enabled, SnapManager creates a dataset for the database. A dataset consists of a collection of storage sets along with configuration information associated with their data. The storage sets associated with a dataset include a primary storage set used to export data to clients, and the set of replicas and archives that exist on other storage sets. Datasets represent exportable user data. If the administrator disables protection for a database, SnapManager deletes the dataset. • If you are using ONTAP, you must select either the *SnapManager_cDOT_Mirror* or *SnapManager_cDOT_Vault* policy depending on the SnapMirror or SnapVault relationship created. When you disable backup protection, a warning message is displayed stating that the dataset will be deleted and restoring or cloning backups for this profile will not be possible.
View the profile	Because the storage administrator has not yet assigned storage resources to implement the protection policy, the profile shows up as nonconformant in both the SnapManager graphical user interface and the profile show command output.
Assign storage resources in the Protection Manager Management Console	In the Protection Manager Management Console, the storage administrator views the unprotected dataset and assigns a resource pool for each node of the dataset that is associated with the profile. The storage administrator then makes sure that secondary volumes are provisioned and protection relationships are initialized.
View the conformant profile in SnapManager	In SnapManager, the database administrator sees that the profile has changed to conformant state in both the graphical user interface and in the profile show command output, indicating that resources were assigned.

If you want to…	Then…
Create the backup	• Select full backup. • Also, select whether the backup should be protected and select the primary retention class (for example, hourly or daily). • If you are using Data ONTAP operating in 7-Mode and want to protect the backup immediately to secondary storage overriding the Protection Manager protection schedule, specify the -protectnow option. • If you are using ONTAP and want to protect the backup immediately to the secondary storage, specify the protect option. The protectnow option is not applicable in clustered Data ONTAP.
View the backup	The new backup is shown as scheduled for protection, but not yet protected (in the SnapManager interface and in the backup show command output). The Protection state is shown as “Not protected”.
View the backup list	After the storage administrator verifies that the backup has been copied to secondary storage, SnapManager changes the backup Protection state from “Not protected” to “Protected”.

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